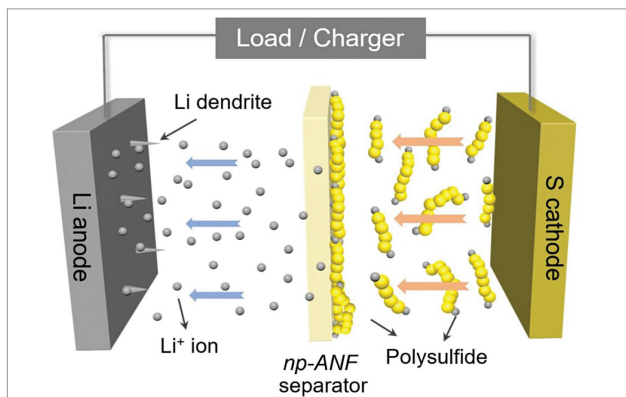


LithiumSulfur battery promises 5 times more capacity

Researchers at Monash University in Melbourne have developed a new battery design that offers numerous benefits over conventional lithium-ion batteries. The team's 'nano-porous polymer-coated lithium foil anode' is a streamlined,



This paper-thin polymer coating is changing the game, and lithium-sulfur batteries could be the future of energy storage (Courtesy: www.arenaev.com)

cost-efficient design that amplifies energy storage. The volatile interaction resembled dramatic soap opera scenes in the battery realm, where lithium-ion counterparts would occasionally flare up. The researchers' ultra-thin polymer coating, peppered with minute holes smaller than a nanometre, acts as a protective barrier. It lets lithium ions flow while preventing harmful chemical interference. The team believes their research can counteract the rapid decay traditionally seen in lithium metal, historically the weak point of lithium-sulphur batteries. As these batteries promise to redefine an energy-hungry market, especially with surging demands in electric vehicles, aircraft, and electronics, it's vital to acknowledge that the battery landscape is ever-evolving.

A battery recycling method harnessing natural acids

Researchers at Chalmers University of Technology in Sweden have efficiently recycled metals from spent electric car batteries. The method recovers 100% aluminium and 98% lithium from batteries. Using oxalic acid, an organic acid found in plants, they minimised the loss of precious raw materials like nickel, cobalt, and manganese. In the experiments, the researchers dissolved pulverised spent car battery contents in oxalic acid, resembling finely ground black powder. The team champions plant-derived oxalic acid over traditional inorganic chemicals used in battery recycling. Conventional methods dissolve all metals in an electric vehicle battery cell, often leading to lithium loss. However, this new approach first recovers lithium and alu-

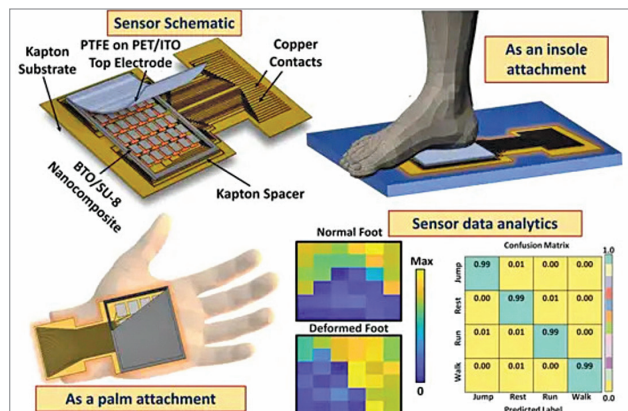


The crushed contents of a spent car battery cell (silver bag), in the form of a finely ground black powder (dish on the left), are dissolved in a transparent liquid—oxalic acid. After a set time, the black mixture is filtered. The aluminium and lithium end up in the (greenish blue) liquid, while the other metals, such as cobalt, nickel, and manganese, remain in the dark 'solids' (dish on the right) (Credit: Chalmers University of Technology, Anna-Lena Lundqvist)

minium, minimising precious metal waste. After filtering, due to the distinct properties of aluminium and lithium, separation becomes uncomplicated.

Wearable pressure sensor for postural deformity monitoring

Researchers at the Indian Institute of Technology Delhi (IIT Delhi) have unveiled a scalable and wearable pressure sensor that revolutionises how one addresses gait and postural deformities. The nanocomposite-based sensor offers a cost-effective alternative to traditional treatments, addressing



Scalable and wearable pressure sensor (Credit: www.educationtimes.com)

issues such as splay foot, flat foot, and related deformities impacting balance, mobility, and joint health. Crafted from a blend of light-sensitive polymers and piezoelectric nanoparticles, this versatile sensor adapts to various deformities, providing vital data for machine learning algorithms and enabling the design of custom insoles. Its versatility extends to detecting different human activities and aiding in rehabilitation, making it a promising tool for various industries, including healthcare, sports, and defence.